Agrovoltaic Canopy for Protection and Sustainable Farming

Six sensors tilt the panels to optimise crop light and electricity; a gutter harvests rain; power is sold to a nearby industry. Digital-twin study, TCE ground, Madurai (9.88 N).

Team: _____
Guide: _____
Dept: _____
R3 to R7

THE PROBLEM AND THE IDEA

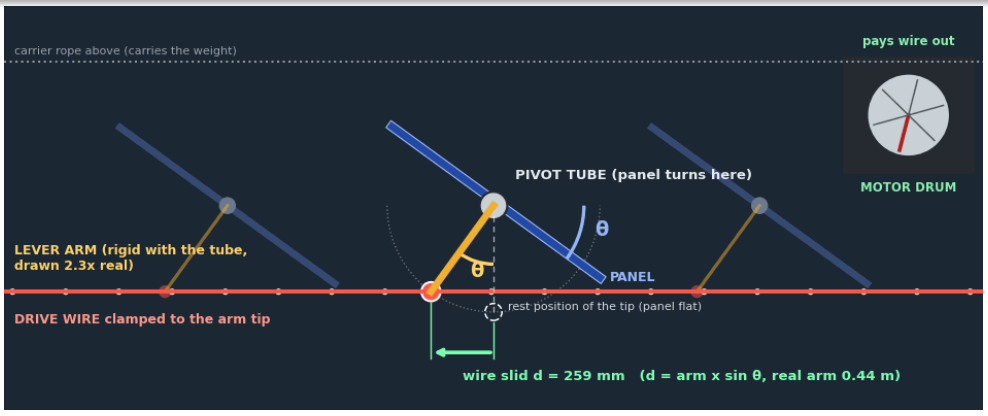
- Small farms live on crop income alone: weather and prices decide.
- One frame over the farm: panels 4.5 m up on stainless wire ropes, crops below.
- Panels tilt for sun. In rain they close into a sealed roof (3 deg fall) that feeds a gutter, an underground cistern and drip irrigation.
- Six sensors tell the controller how to balance electricity and crop light.
- The power is sold to a nearby industry: the farmer gets a monthly share plus crop income.

THE DESIGN (BLENDER RENDER, TCE GROUND)



10 panels on a 25 m wire run; pivot 5.58 m; lowest edge 4.5 m (harvester fits)

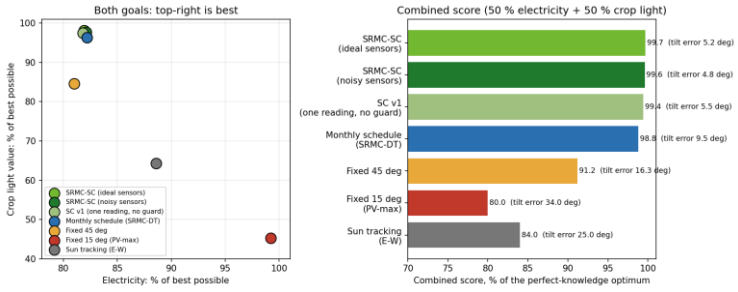
HOW THE MOTOR TILTS THE PANELS



- Wire slides by d; it is clamped to the tip of a lever arm on every panel.
- The arm turns the pivot tube: tilt = arm swing, d = arm x sin(tilt).
- A guide pulley lets the wire turn at the row end.

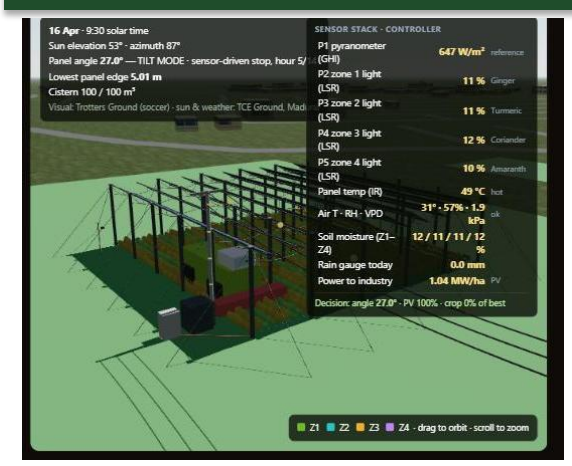
THE SIX-SENSOR CONTROLLER

- Six sensors: pyranometer, PAR, DHT11, soil moisture, rain, wind. Rain = flat roof (3 deg); night = park.
- Otherwise: best tilt 15-60 deg for 50 % electricity + 50 % crop light. PAR corrects the model; heat or dry soil = more shade.
- Year-long test: 99.6 % of the best score (electricity 82 %, crop light 98 %).



Both goals (top-right is best): six-sensor controller 99.6 % of the best score

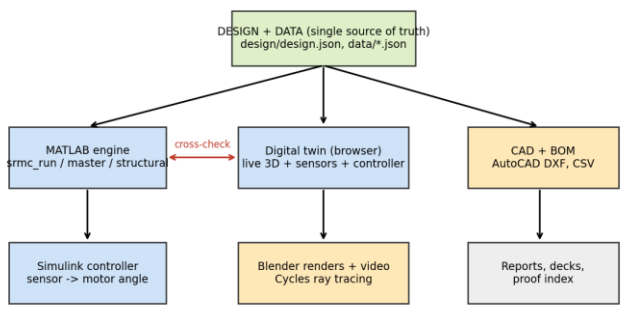
LIVE DIGITAL TWIN (OFFLINE, IN A BROWSER)



- Sun, shadows, panel angle, drive wire, sensors, water, money.
- 7 climates, 9 soils, 24 crops.

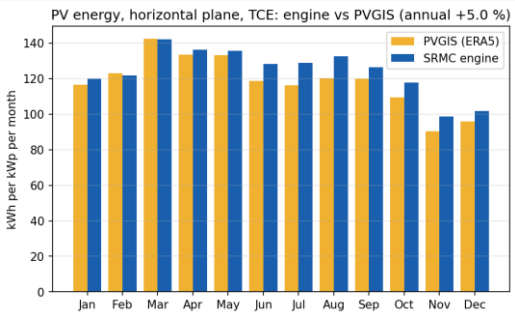
⚠ Sensors are simulated, not measured.

ONE DESIGN FILE FEEDS EVERYTHING

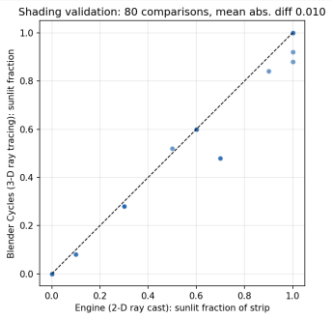


- MATLAB + Simulink, web twin, AutoCAD DXF + 22-item BOM, Blender Cycles. MATLAB and twin agree to 0.0000 % on 22 outputs.

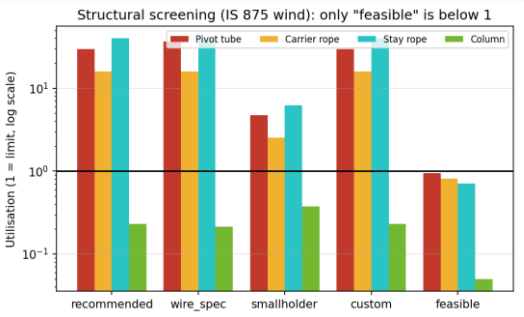
PROOF: THREE INDEPENDENT CHECKS



Solar output vs PVGIS: +5.0 % (sunshine data differs by +5.2 %)



Shadows vs Blender 3-D ray tracing: mean difference 0.010 (80 cases)



Wind check (IS 875): 12.5 m bays fail (rope 16x); 5 m rope grid passes

RESULTS PER HECTARE (TCE, ASSUMED PRICES)

Design	PV kWh/m2	Crop light	Total INR lakh
Hourly plan (SRMC-DT)	238	34 %	55.0
Fixed 45 deg	233	32 %	52.5
PV-only, fixed 15 deg	291	14 %	68.2

- Farmer income about INR 2 lakh per ha per month (range 1.4 to 3.0).
- Rain covers 94 % of irrigation.
- Controller: +55 % crop income vs fixed 45 deg.

HONEST LIMITS

⚠ Simulation only: no field or prototype data.

⚠ Tariff (INR 3/kWh) and farmer share (30 %) are assumptions.

⚠ Investor payback is long (about 41 years).

- A PV-only design earns more money; the controller case rests on crops, water and soil.
- Structure is a screening check, not a certified design.

Data: NASA POWER, PVGIS (EC-JRC), OpenStreetMap (ODbL). Tools: MATLAB / Simulink R2025a, Blender 5.2, three.js. Full R3-R7 decks, report, viva Q&A and proof index in the review pack.